

WAYNE WU

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SUMMARY

Software engineer building production systems across computer graphics, machine learning, and scalable pipelines, bridging research and production for large-scale 3D content generation.

SKILLS

Languages: C++, Python, CUDA, GLSL/HLSL, JavaScript

Graphics: Houdini, Unity, Vulkan, WebGPU, Omniverse

ML & Data: PyTorch, ComfyUI, MediaPipe

Systems: AWS, Distributed Pipelines

PROFESSIONAL EXPERIENCE

Senior Graphics Engineer (Technical Lead), Genies (Los Angeles, CA)

Sep 2023 — Dec 2025

- **UGC Asset Processing Pipeline:** Architected and launched a 0→1 cloud-native asset processing pipeline for 5,000+ AI-generated 3D avatars and wearables, defining system architecture across graphics libraries, asset pipeline, ML model integration, and job orchestration.
- Built per-asset cost models (~\$0.40/asset baseline) for GPU-backed pipelines on AWS, driving architectural optimizations (pipeline simplification, workload decomposition) that reduced latency by over 60%
- **Avatar Auto-Rigging System:** Led development of a UGC auto-rigging system for non-templated avatars and wearables using geometry processing, procedural rigging, simulation, and image-based diffusion models, enabling scalable ingestion and processing of diverse user-generated assets.
- **OpenUSD Processing Framework:** Designed a scalable OpenUSD-based framework leveraging vectorized computation, structured avatar data schemas, and task orchestration for iterative processing to support high-throughput auto-rigging pipelines; presented at SIGGRAPH 2025.
- **Synthetic Data Generation in Houdini:** Defined procedural randomization and labeling workflows, enabling scalable creation of labeled 2D/3D character datasets for ML training and evaluation.

Lecturer (GPU Programming), University of Pennsylvania (Philadelphia, PA)

Sep 2023 — Dec 2023

- Designed and taught graduate-level coursework for 30 students on advanced GPU algorithms, frameworks, and architecture, covering the latest features in CUDA, Vulkan, and WebGPU.

R&D Engineer (Production), Industrial Light & Magic (San Francisco, CA)

Jun 2021 — Sep 2021

- **Procedural USD Workflows:** Applied procedural workflows to ILM's pipeline, shaping the studio's long-term USD strategy across departments and presented the work at SIGGRAPH 2021.

Technical Director (Production Technology), Blue Sky Studios (Greenwich, CT)

Aug 2019 — Jan 2021

- **Asset Pipeline Integration in Houdini:** Integrated the studio's asset pipeline into Houdini as node-based workflows enabling large-scale asset processing (30,000+ show assets) by technical artists.
- **USD Workflows for FX:** Established workflows in OpenUSD and Solaris (VEX, Python) for simulation, data translation, and rendering pipelines, improving artists iteration speed and production reliability.
- **Houdini & USD Training:** Trained 50+ artists and TDs on pipeline, USD, Solaris, and PDG concepts, and interfaced with lead artists, supervisors, and external vendors in formulating workflow design options.

3D Software Developer (Co-op), SideFX (Toronto, ON)

Fall 2017 & Summer 2018

- **RBD Material Fracturing:** Developed new procedural fracturing techniques for rigid-body simulation with modular artistic controls and threaded execution for 4x performance.

ACADEMIC EXPERIENCE

University of Pennsylvania, Philadelphia, PA, USA

Master of Science in Engineering, **Computer Graphics and Game Technology**, University of Pennsylvania, 2023

- **Wiggly (Authoring Tool):** Implemented an optimization-based algorithm for animating deformable objects in Houdini (C++), and designed a modular and procedural workflow (Python, VEX) for artists to easily create animations.
- **Real-time Crowd Simulation:** Developed the first WebGPU-based interactive crowd simulation using position-based dynamics, compute shaders (WGSL, JS) and optimized render pipelines to support 10,000+ agents in real time.

University of Waterloo, Waterloo, ON, Canada

Bachelor of Applied Science (Honors), **Systems Design Engineering**, 2019

- Graduated with Distinction, 2019 | Engineering Faculty/Staff Upper Year Scholarship, 2018